Post-Quantum DNSSEC on the Wire: Measuring Transport Frontiers in TurboDNS and SL-DNSSEC

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***Abstract*—Post-quantum DNSSEC makes DNS transport a first-order systems constraint because larger signatures and DNSKEY material push validating resolution through EDNS sizing, truncation, TCP fallback, retransmission, and cache state. We compare two recent proposals with different execution paths: TurboDNS, which preserves signed DNSSEC and accelerates TCP fallback, and SL-DNSSEC, which validates steady-state answers through a KEM-and-MAC construction with DNSKEY bootstrap support. Both public artifacts are run in the same validating DNSSEC environment, with authoritative and recursive probes used to collect repeated executions and packet captures across network profiles, EDNS sizes, controlled response-size ladders, cache phases, and signature families.**

The measurements identify a transport frontier. In the core Falcon-512 benchmark, EDNS 4096 moves TurboDNS authoritative DNSKEY delivery into DNS-layer UDP completion, while SL-DNSSEC remains truncated on that authoritative DNSKEY path. The same split persists for inflated signed `A` answers: under Falcon-512, TurboDNS keeps the EDNS-4096 authoritative window through `bulk128.example` and loses it at `bulk192.example`, whereas the evaluated SL-DNSSEC artifact lacks sustained validating completion in those large-answer rows. Resolver-side packet captures show the residual end-to-end cost behind local authoritative gains, including TCP frames and retransmissions, and trust-material warmup improves TurboDNS robustness near the large-answer boundary. A signature-threshold micro-study further bounds the result by showing that the Falcon-512 window occupies a non-empty tested region, while the tested Dilithium2 range is truncated from the first point.

The results show that EDNS size, effective response size, network profile, cache phase, and signature family jointly determine the transport regime induced by post-quantum DNSSEC designs.

I. Introduction

DNSSEC already pushes ordinary DNS transport toward large responses, truncation, and fallback, and post-quantum signatures increase that pressure by enlarging both key material and signed answer material. Large DNSKEY RRsets, large signed answer RRsets, and repeated validation traffic tie EDNS sizing to fragmentation risk, retransmission, and cache state, so the operational behavior of a design depends on the transport regime induced by its primitive inside a validating DNSSEC path [1]–[4].

Two recent proposals address this pressure through different execution paths. TurboDNS preserves the signed DNSSEC model and reduces the cost of TCP fallback by embedding part of the TCP setup into the initial DNS/UDP exchange before streaming the full answer over TCP [5]. SL-DNSSEC changes the steady-state answer-validation path more substantially by using a post-quantum KEM and a MAC for ordinary answer validation, while retaining a DNSKEY bootstrap path in the background [6], [7]. Both proposals target the same operational bottleneck, but they place the resulting transport burden on different parts of the validating path.

Comparing these designs requires separating the planes on which the cost is paid. A design can perform well at the authoritative server while shifting work to the validating resolver, reduce bytes in one phase while increasing fallback pressure in another, or benefit from a warm trust path while exposing different behavior on a cold validating path. With these planes separated, the systems question becomes concrete: where each design completes in UDP, where it pays TCP, and under which conditions one regime remains preferable to the other.

We answer that question through a unified comparative execution of the public TurboDNS and SL-DNSSEC artifacts. Both prototypes are aligned in the same four-node validating DNSSEC environment and instrumented with authoritative and recursive probes. We analyze repeated measurements together with packet captures across network profiles, EDNS sizes, response-size ladders, cache phases, and signature families, locating the frontier at which each design keeps a local advantage, loses that advantage, or carries it through the end-to-end validating path.

The evaluation yields five technical contributions.

1) A transport-centered comparison of post-quantum DNSSEC that distinguishes authoritative completion, the first resolver-visible leg, and end-to-end validating completion.
2) A unified experimental methodology that runs TurboDNS and SL-DNSSEC in the same validating environment under the same network profiles, EDNS settings, response-size ladders, and packet-capture instrumentation.
3) The core frontier over network profile and EDNS size. In the core benchmark, EDNS 4096 is the only setting that produces DNS-layer authoritative UDP-complete DNSKEY delivery for TurboDNS under Falcon-512, while SL-DNSSEC keeps the authoritative DNSKEY path truncated throughout.
4) Evidence that the structural split survives inflated signed `A` answers, localizes the Falcon-512 high-size boundary between `bulk128.example` and `bulk192.example`, and isolates the effect of trust-material warmup on large-

answer validation.

5) Evidence that the local EDNS-4096 window of TurboDNS is signature-family dependent: it is present for Falcon-512 in a non-empty region and absent in the tested Dilithium2 range.

II. Background and Related Work

DNSSEC authenticates DNS data with DNSKEY, DS, RRSIG, and authenticated denial records [8]–[10]. In operational terms, DNSSEC already stresses transport in the classical setting: larger responses increase truncation, trigger TCP fallback, and make EDNS sizing consequential. These failure modes have been documented repeatedly in the DNS measurement literature, including work on fragmentation-induced unreachability and large-message failure modes [2], [3], [11]. Post-quantum signatures intensify the same pressure because both DNSKEY material and signed answer RRsets grow substantially [4].

Recent post-quantum DNSSEC work has followed three broad directions. One direction keeps signature-based validation and studies how to carry large post-quantum objects over UDP, for example through request-based or QNAME-based fragmentation schemes [12], [13]. A second direction optimizes the fallback path. TurboDNS belongs to this family: it preserves ordinary signed DNSSEC validation and lowers the effective cost of TCP fallback by moving handshake material into the initial DNS/UDP exchange before streaming the response over TCP [5]. A third direction changes the answer-validation primitive itself. SL-DNSSEC belongs here: it uses a post-quantum KEM and a MAC to validate ordinary responses while retaining a DNSKEY bootstrap path [6]. The subsequent security analysis formalizes the cryptographic core of that construction; the placement of transport cost across a validating path remains a systems question [7].

This paper studies that systems question through a unified execution of the public TurboDNS and SL-DNSSEC artifacts in one validating environment. The evaluation identifies where the transport burden appears, where it disappears, and how that answer changes with EDNS size, response size, cache phase, network profile, and signature family. The contribution is a systems comparison with new experimental evidence, focused on the transport consequences of post-quantum DNSSEC designs.

This positioning separates our contribution from the individual papers on TurboDNS and SL-DNSSEC. Those papers establish the designs and report their own controlled evaluations. The comparison here places both systems under one validating harness, uses repeated packet-level evidence across the same network profiles and response-size ladders, and evaluates whether local authoritative transport behavior survives as an end-to-end validating advantage. The present paper is a comparative systems study of the transport consequences of those designs.

To make that comparison precise, we distinguish three observation planes. The *authoritative path* asks whether the authoritative server can complete a response in UDP under the advertised EDNS budget. The *first resolver-visible leg* asks what a validating resolver first exposes on its outward-facing side for the query under study. The *validating path* is the full completion path for that query, including the control and validation traffic that the resolver must pay internally. This separation exposes cases where a design completes locally in UDP and still pays substantial TCP work end-to-end.

We also treat transport outcomes as categorical observations derived from probe windows and packet captures. For a system S under a cell c and a query class q, we report a success rate

$$P_{\mathrm{succ}}(S,c,q) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[\text{NOERROR and AD on run } i],$$

with n repeated executions. We summarize successful validating-query latencies with the median and $p95$, and we classify transport windows into regimes such as `udp_complete`, `udp_truncated`, and `udp_and_tcp`. The labels are direct summaries of packet-level observations in the corresponding probe windows.

III. Experimental Methodology

A. Execution environment

We evaluate both systems in the same four-node validating DNSSEC topology: a root nameserver, an authoritative child nameserver for `example`, a validating resolver, and a client. The topology follows the structure used by the released artifacts and places both systems in one common execution environment. For comparability, we normalized the released artifacts so that both systems could be built, started, and measured under the same runners, naming layout, and probe discipline. The normalization preserves the published protocol roles and evaluation intent and removes startup and configuration faults unrelated to the transport question studied here. Protocol message formats, cryptographic validation logic, and fallback mechanisms remain as published.

The environment exposes two workload families. The baseline family exercises the validating path centered on `test0.example`and the accompanying DNSKEY probes. The inflated-answer family uses signed `A` responses, where `bulkNN.example` denotes an owner name with N signed `A` records under the same name. This keeps the query type fixed while varying effective response size. The resulting ladder is a controlled payload experiment; its RRset cardinalities are used to stress response size under a fixed query type. Figure 2 summarizes the common topology and the two workload families used across all campaigns.

B. Network profiles, EDNS, and repeated execution

The core campaign spans three network profiles, denoted *clean*, *mid*, and *harsh*. These profiles expose both designs to progressively more difficult transport conditions under the same controlled parameters. Table II gives the concrete delay, jitter, loss, and rate parameters. Link MTU remains 1500 B except in the dedicated MTU branch. Benchmark queries use `dig +timeout=8 +tries=1`. The validating resolver keeps `resolver-query-timeout 30000` and

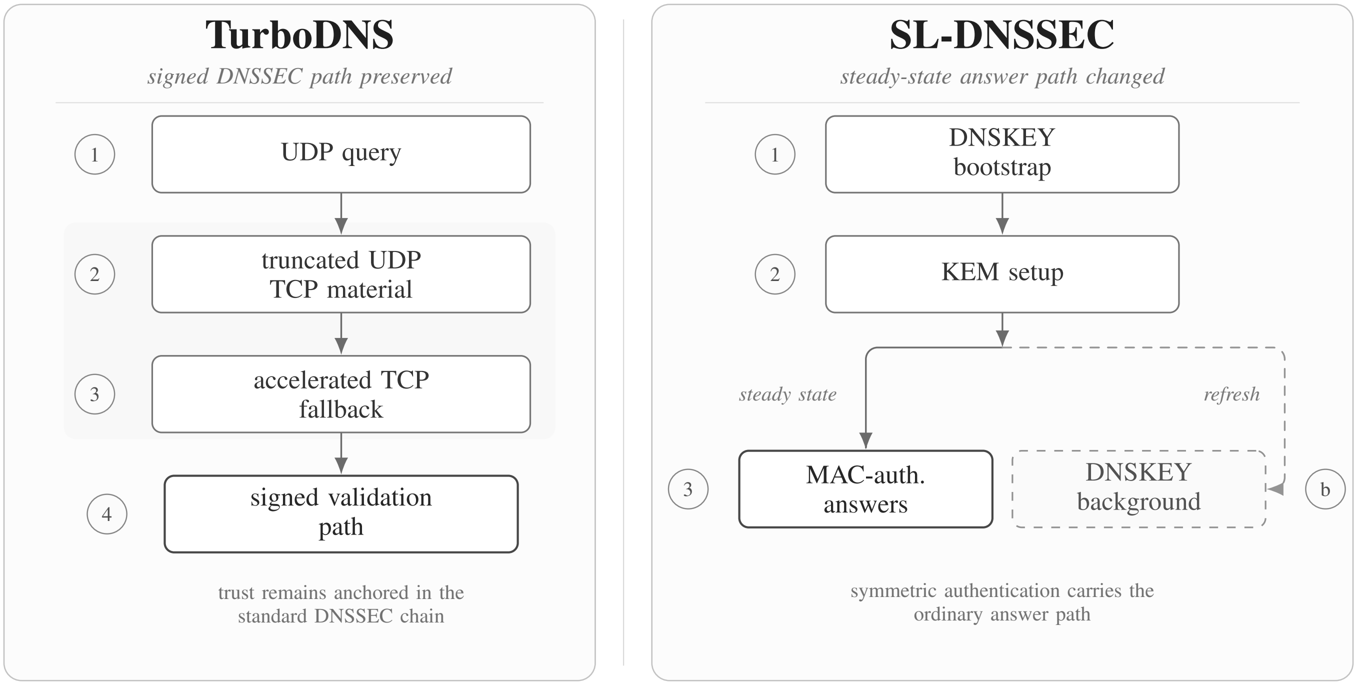

Fig. 1. Comparative overview of TurboDNS and SL-DNSSEC. TurboDNS preserves the signed DNSSEC model and shifts effort into fallback. SL-DNSSEC changes the steady-state answer-validation path while retaining a DNSKEY bootstrap path.

TABLE I

CAMPAIGN FAMILIES USED IN THE PAPER. THE FIRST THREE ROWS DEFINE THE CENTRAL EVIDENCE. THE REMAINING ROWS SHARPEN SPECIFIC QUESTIONS ABOUT STRESS BEHAVIOR, HIGH-SIZE BOUNDARIES, CACHE PHASE, AND SIGNATURE SENSITIVITY.

Campaign	Networks	EDNS	Workload	Repeats	Purpose
Core frontier	clean, mid, harsh	1232, 1472, 4096	`test0.example`+ DNSKEY	30	central transport frontier
Size sweep	clean, harsh	1232, 1472, 4096	`bulk01.example`–`bulk16.example`	8	inflated-`A` behavior below boundary
High-size boundary	clean, harsh	1472, 4096	`bulk32.example`, `bulk96.example`, `bulk128.example`, `bulk192.example`	8	location of Falcon EDNS-4096 window
Cache phase	clean, harsh	1232, 4096	`bulk128.example`, `bulk192.example`	6	cold versus trust-warm comparison
Signature micro	clean	4096	`test0.example`, `bulk01.example`, `bulk04.example`, `bulk08.example`	4	Falcon versus Dilithium2 threshold contrast

TABLE II

CONCRETE NETWORK-PROFILE PARAMETERS USED. DELAY, JITTER, RATE, AND LOSS COME DIRECTLY FROM THE APPLIED NETEM PROFILE. PACKET ORDER FOLLOWS THE DEFAULT NETEM CONFIGURATION.

Profile	Delay	Jitter	Loss	Rate
clean	10 ms	0 ms	0.0%	100 Mbit/s
mid	25 ms	5 ms	0.3%	20 Mbit/s
harsh	50 ms	15 ms	1.0%	5 Mbit/s

`resolver-nonbackoff-tries 1` in both artifacts. The central EDNS settings are 1232, 1472, and 4096 bytes. We also ran a targeted 512-byte stress campaign and a smaller MTU sensitivity study; those results are reported in the appendix.

The main core campaign uses 30 repeated executions per cell. Additional focused campaigns use 6 to 8 repetitions per cell, depending on purpose. Repetition captures the tails and intermittent failures that one-shot runs can hide in post-quantum DNSSEC transport. The core campaign identifies the stable frontier over network profile and EDNS size. The focused campaigns then clarify three questions: where the Falcon EDNS-4096 window disappears for larger signed `A` answers, how trust-material warmup changes the validating path for those large answers, and how much of the Falcon result survives when the signature family changes.

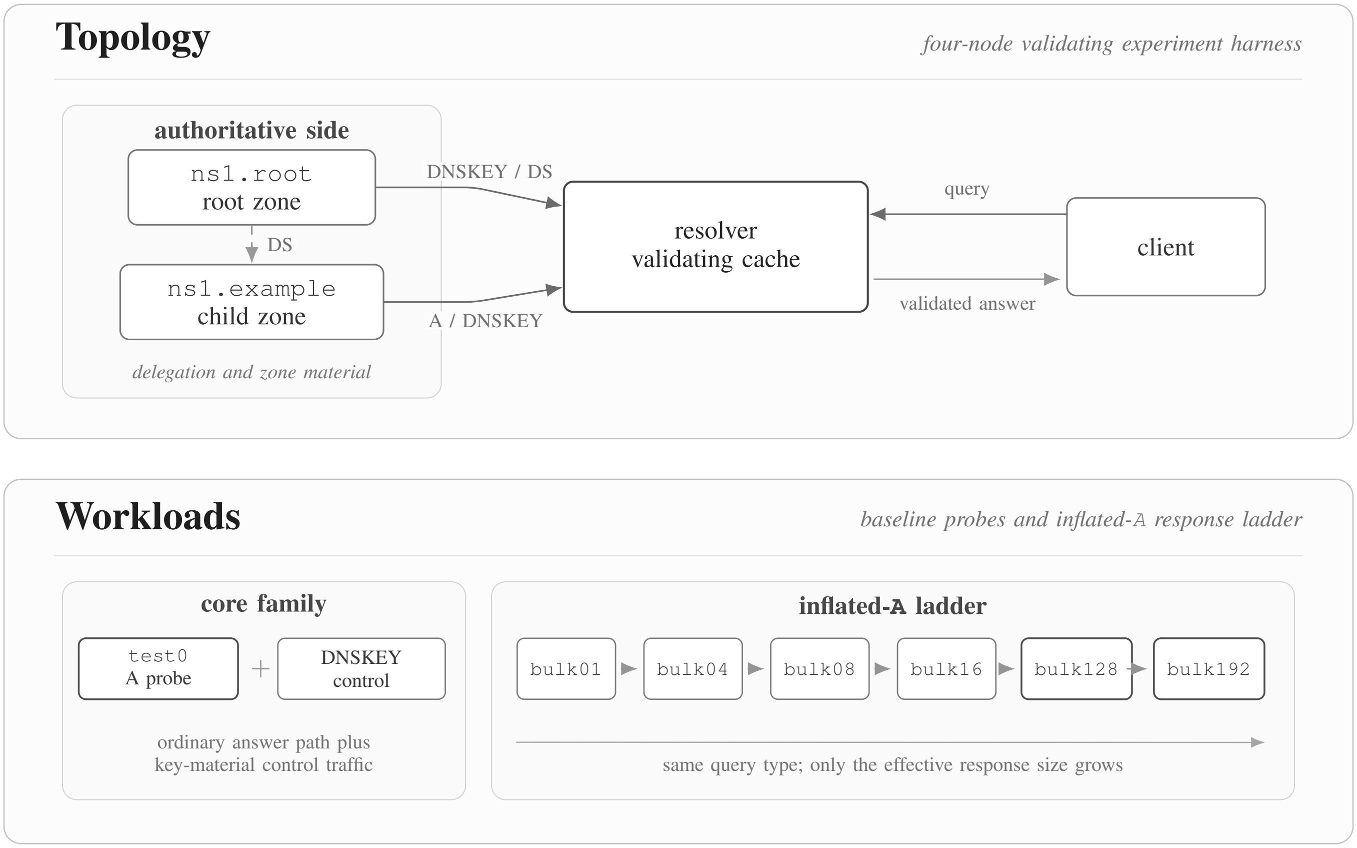

Fig. 2. Common four-node validating topology and the two workload families. The core family centers on `test0.example` and DNSKEY probes, while the inflated-`A` ladder keeps the query type fixed and grows only the effective response size.

C. Measurement planes and transport instrumentation

Each cell is evaluated with direct authoritative probes, recursive probes, repeated validating queries, and packet captures. The probe suite gives three views of the system: authoritative completion, first resolver-visible behavior, and full validating completion. Packet captures are segmented into probe windows so that transport summaries are tied to specific queries.

For each system, cell, and query class we report:

- success rate P_{succ} for validating completion,
- median validating-query latency over successful runs,
- the authoritative transport regime,
- the first resolver-visible regime,
- packet-level totals for bytes, DNS frames, TCP frames, truncation, and retransmissions.

The measurements separate local authoritative behavior from full validating cost. A local authoritative UDP advantage can coexist with substantial TCP work in the end-to-end validating path. Packet captures expose that split. The probe-window exports attribute bytes, DNS frames, TCP frames, truncation, and retransmissions to specific measurements, which allows the local authoritative gain to be compared with the cost that remains in the full validating path. The artifact exports retain per-run series and $p95$ summaries, while the main paper tables emphasize medians for compactness.

Labels such as `udp_complete` or `UDP` denote DNS-layer completion: a UDP reply with TC=0 and no TCP completion in the corresponding probe window. They should not be read as fragmentation-free delivery at the IP layer. Conversely, `udp_truncated` or `trunc.` denotes a UDP reply with TC=1, `udp_and_tcp` denotes truncation followed by TCP completion in the same window, `mixed` denotes that no single regime dominates the repeated executions for that cell, `servfail` denotes resolver-visible SERVFAIL on the first visible leg, and `fail/trunc.` denotes rows in which the observed successful and unsuccessful runs do not collapse to one packet-window label.

In the isolated cache-phase branch, the *cold* and *trust-warm* phases are executed on separate fresh stacks. The trust-warm phase performs one TCP `example. DNSKEY` query and one TCP `example. DS` query before the measured validating `A` query, while the control probes run after the benchmark query so that the cold phase remains cold.

IV. Core Frontier over Network Profile and EDNS

Table III summarizes the central benchmark over *clean*, *mid*, and *harsh* network profiles and EDNS values 1232, 1472, and 4096.

a) EDNS 4096 is the only core setting that changes the authoritative DNSKEY regime.: Under Falcon-512, TurboDNS

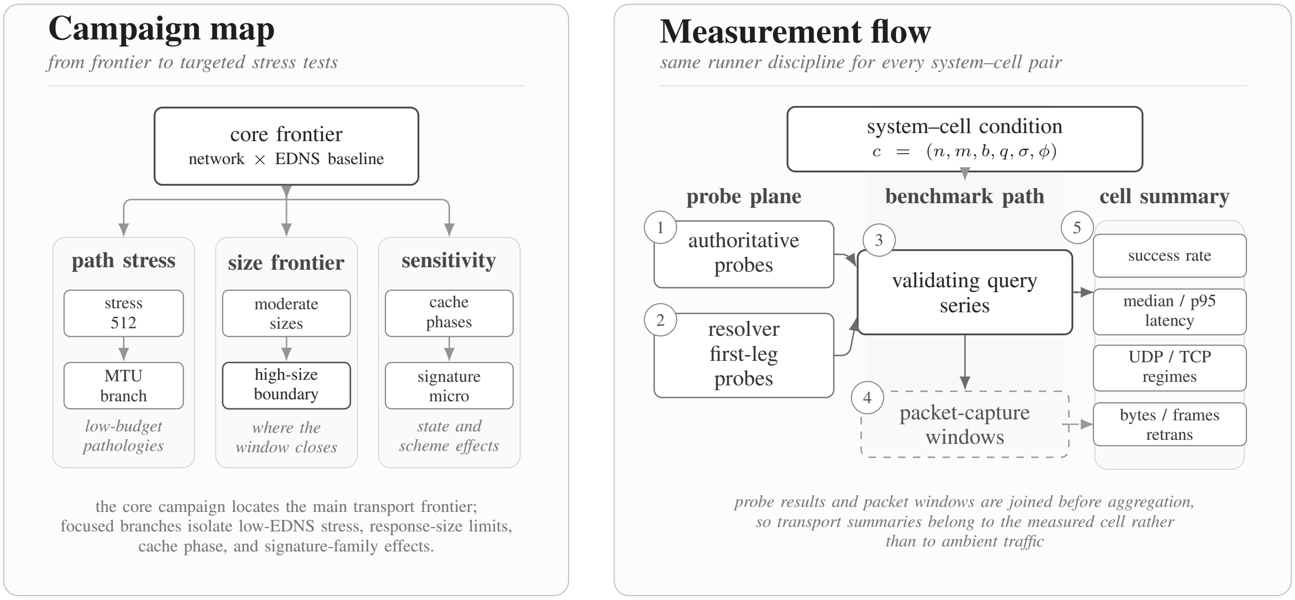

Fig. 3. Campaign organization and per-cell measurement flow. The left panel summarizes the benchmark family tree; the right panel shows how each cell combines direct probes, repeated validating queries, packet-capture windows, and summary statistics.

TABLE III

Core transport frontier. Success rates are measured over 30 repeated executions per cell. The authoritative DNSKEY and first-leg columns report the dominant observed DNS-layer transport regimes. Reported validating medians are conditional on successful runs only.

Net.	EDNS	SL succ.	Turbo succ.	Joint succ.	SL auth DNSKEY	Turbo auth DNSKEY	SL first leg	Turbo first leg	Med. A (SL/T) ms
clean	1232	0.967	0.967	0.933	trunc.	trunc.	UDP	trunc.	158 / 140
clean	1472	0.967	0.967	0.933	trunc.	trunc.	UDP	trunc.	2999 / 1079
clean	4096	0.933	1.000	0.933	trunc.	UDP	UDP	trunc.	64.5 / 105
mid	1232	0.933	0.967	0.900	trunc.	trunc.	UDP	trunc.	123.5 / 225.5
mid	1472	0.867	0.967	0.833	mixed	trunc.	UDP	trunc.	2999 / 2445
mid	4096	0.900	0.933	0.833	trunc.	mixed	UDP	mixed	121.5 / 228
harsh	1232	0.900	0.900	0.800	mixed	trunc.	UDP	mixed	250.5 / 474
harsh	1472	0.733	0.800	0.600	mixed	mixed	mixed	trunc.	2105 / 479.5
harsh	4096	0.833	0.900	0.767	trunc.	UDP	UDP	fail/trunc.	235 / 1551

reaches authoritative DNS-layer UDP-complete DNSKEY delivery at EDNS 4096. SL-DNSSEC keeps the authoritative DNSKEY path truncated in all core rows. EDNS 1472 remains in the same truncated authoritative DNSKEY regime as 1232. The two systems therefore induce distinct transport regimes at the authoritative server.

b) The user-facing first resolver leg remains structurally different.: Across the core rows, SL-DNSSEC typically exposes a UDP-complete first visible leg for the validating `A` query, while TurboDNS typically exposes a truncated first leg. This behavior is stable across network profiles and reflects a structural property of the designs.

c) A local authoritative gain leaves residual end-to-end cost.: The packet-level summaries expose the remaining validating-path burden. In the resolver path, TurboDNS can retain more TCP frames and more retransmissions even when it has a local authoritative advantage. For example, in the EDNS-4096 resolver path under *clean*, both SL-DNSSEC and TurboDNS complete successfully, but TurboDNS still carries a larger TCP burden in the validating path. The same pattern appears under *harsh*. The comparison must therefore include both authoritative behavior and resolver-side completion cost.

d) The frontier is probabilistic.: The core campaign uses repeated executions to capture tails and intermittent failures. In the *clean* rows, success rates are high for both systems, with residual failures. In *mid* and *harsh*, the gap between median behavior and tail behavior widens. The *harsh*/1472 row is the least stable row of the core benchmark and marks the region where the transport regime becomes fragile.

Across the nine core rows, TurboDNS never falls below SL-DNSSEC in core success rate and exceeds it in several configurations. The latency ordering depends on configuration: under EDNS 4096, SL-DNSSEC often has the lower median validating latency, whereas under EDNS 1472, TurboDNS can have the lower median. Since these medians are conditioned on successful validating completion, rows with different success

TABLE IV
REPRESENTATIVE PACKET-LEVEL COST SUMMARIES FROM THE RESOLVER-SIDE CAPTURE WINDOWS FOR EDNS 4096. VALUES ARE MEDIANS OVER SUCCESSFUL RUNS IN THE CORE BENCHMARK.

Net.	System	Bytes	DNS frames	TCP frames	TCP retrans.
clean	SL-DNSSEC	72.5k	148	34	8.5
clean	TurboDNS	85.9k	145	84	37.5
harsh	SL-DNSSEC	79.2k	158.5	33	12.5
harsh	TurboDNS	83.5k	146.5	82.5	38.0

rates must be read together with their corresponding success probabilities. The comparison is therefore reported as a frontier over EDNS size, completion probability, and validating latency.

e) Resolver-side captures expose the remaining validating-path cost.: Table IV reports representative resolver-side packet-level summaries for EDNS 4096. Under *clean*, SL-DNSSEC completes the validating path with a median of roughly 72.5 kB on wire, 34 TCP frames, and 8.5 TCP retransmissions, whereas TurboDNS uses about 85.9 kB, 84 TCP frames, and 37.5 retransmissions. Under *harsh*, the same pattern remains: SL-DNSSEC uses around 79.2 kB with 33 TCP frames and 12.5 retransmissions, whereas TurboDNS uses about 83.5 kB with 82.5 TCP frames and 38 retransmissions. The local authoritative advantage of TurboDNS at EDNS 4096 is real, and the resolver-side captures show the TCP cost that remains in the full validating path.

The decisive variables are EDNS size, effective response size, network profile, and the observation point within the validating path. The next sections show that the same structural split survives larger signed `A` answers, cache effects, and changes in signature family.

V. RESPONSE SIZE, CACHE PHASE, AND SIGNATURE FAMILY

A. The split persists for inflated signed `A` answers

The first size sweep evaluates whether the transport split observed for DNSKEY also appears for ordinary signed `A` answers when those answers are inflated in a controlled way. For `bulk01.example`, `bulk04.example`, `bulk08.example`, and `bulk16.example`, SL-DNSSEC keeps authoritative `A` delivery truncated under all EDNS values, while TurboDNS reaches authoritative DNS-layer UDP-complete delivery at EDNS 4096. At the same time, SL-DNSSEC keeps a UDP-complete first resolver-visible leg and TurboDNS keeps a truncated one. The same structural pattern therefore appears with a fixed `A` query type and a controlled increase in effective response size.

B. Falcon-512 has a bounded high-size window

The high-size boundary campaign tightens that observation around the regime where the local EDNS-4096 advantage of TurboDNS disappears. Table V shows the result. Under Falcon-512, TurboDNS authoritative delivery remains DNS-layer UDP-complete through `bulk128.example` and disappears at `bulk192.example`. In the evaluated SL-DNSSEC artifact and configuration, the same large-answer rows do not sustain validating completion.

This boundary gives a concrete operating range for the EDNS-4096 benefit: it identifies the response sizes and signature family for which the authoritative path remains non-truncated at the DNS layer. Here and throughout, DNS-layer UDP-complete means a non-truncated DNS reply at the advertised EDNS size; it does not imply fragmentation-free delivery at the IP layer.

C. Trust-material warmup changes TurboDNS in the large-answer regime

The cache-phase experiment isolates cold and trust-warm execution on fresh stacks for `bulk128.example` and `bulk192.example`. The experiment separates first-access behavior from runs in which trust material has been warmed before the measured validating query. Table VI reports representative rows.

The result is strong and asymmetric. For TurboDNS, trust warmup often converts a partially successful cold regime into a more reliable validating path for large signed `A` answers. The improvement is most visible in success rate. Latency does not follow a uniform monotone pattern: some warm rows become much faster, while others remain expensive even as success improves. Trust-material warmup is therefore best read as a robustness intervention. For SL-DNSSEC, the measured large-answer rows remain without sustained validating completion. The experiment separates same-query cache behavior from a broader trust-path effect by injecting the warmup before the measured query on a fresh stack.

D. The local EDNS-4096 window depends on the signature family

The final micro-study evaluates whether the local EDNS-4096 window follows from the fallback design alone or from its interaction with the signature family carried by the authoritative path. The comparison combines two available evidence branches: the clean/4096 Falcon size-sweep cells ($R = 8$) and the dedicated clean/4096 Dilithium2 threshold micro ($R = 4$). Under Falcon-512, TurboDNS retains authoritative DNS-layer UDP-complete delivery at EDNS 4096 for `bulk01.example`, `bulk04.example`, and `bulk08.example`; the accompanying moderate size sweep also stays non-truncated at `bulk16.example`. Under Dilithium2, the local authoritative advantage is absent already at `test0.example`and remains absent through `bulk08.example`. At the same time, the structural resolver-side split remains visible: SL-DNSSEC continues to expose a UDP-complete first leg, and TurboDNS continues to expose a truncated one. The Falcon window is therefore an operating region produced by the interaction between the fallback design and the signature family that still populates the signed path.

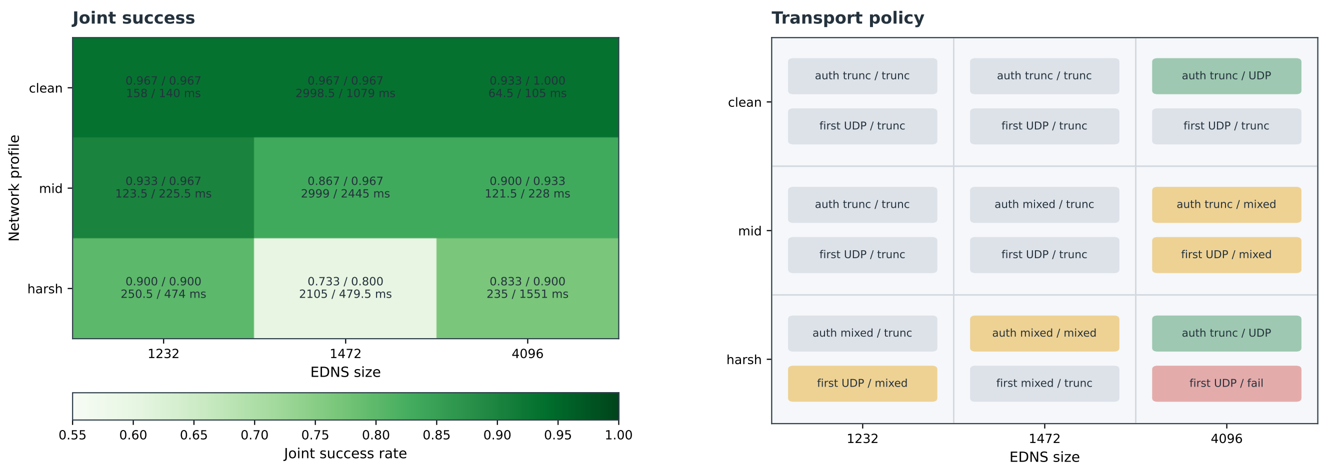

Fig. 4. Two-panel summary of the core benchmark. The left panel reports joint success together with per-system rates and validating medians. The right panel shows the dominant authoritative DNSKEY regime and the first resolver-visible regime in each cell.

TABLE V

HIGH-SIZE BOUNDARY AND SIGNATURE-FAMILY SENSITIVITY. FALCON ROWS REPORT THE OBSERVED SUCCESS COUNTS AND DNS-LAYER AUTHORITATIVE REGIMES FROM THE FALCON BOUNDARY AND MODERATE SIZE-SWEEP BRANCHES; DILITHIUM2 ROWS REPORT THE DEDICATED THRESHOLD MICRO. RETURNED UDP BYTES IN TC-ONLY ROWS ARE TRUNCATED REPLIES.

Branch	Net.	Qname	SL x/R	Turbo x/R	Turbo @1472	Turbo @4096	Returned UDP msg @4096
Falcon boundary	clean	`bulk128.example`	0/8	8/8	trunc.	UDP	3626 (full)
Falcon boundary	clean	`bulk192.example`	0/8	7/8	trunc.	trunc.	72 (TC-only)
Falcon boundary	harsh	`bulk128.example`	0/8	7/8	trunc.	UDP	3626 (full)
Falcon boundary	harsh	`bulk192.example`	0/8	8/8	trunc.	trunc.	72 (TC-only)
Falcon sweep	clean	`bulk01.example`	7/8	8/8	–	UDP	2307 (full)
Falcon sweep	clean	`bulk04.example`	8/8	8/8	–	UDP	2355 (full)
Falcon sweep	clean	`bulk08.example`	8/8	8/8	–	UDP	2419 (full)
D2 micro	clean	`test0.example`	4/4	4/4	–	trunc.	70 (TC-only)
D2 micro	clean	`bulk01.example`	4/4	4/4	–	trunc.	71 (TC-only)
D2 micro	clean	`bulk04.example`	4/4	4/4	–	trunc.	71 (TC-only)
D2 micro	clean	`bulk08.example`	4/4	4/4	–	trunc.	71 (TC-only)

TABLE VI

REPRESENTATIVE ROWS FROM THE ISOLATED CACHE-PHASE EXPERIMENT. COUNTS REPORT SUCCESSES OVER REPEATED FRESH-STACK EXECUTIONS. TRUST-MATERIAL WARMUP IMPROVES TURBODNS ROBUSTNESS IN SEVERAL LARGE-ANSWER ROWS, WHILE SL-DNSSEC SHOWS NO SUSTAINED VALIDATING COMPLETION IN THE SAME REGIME. MEDIANS ARE CONDITIONAL ON SUCCESSFUL RUNS.

Net.	EDNS	Qname	Phase	SL x/R	Turbo x/R
clean	1232	`bulk128.example`	cold	0/6	4/6
clean	1232	`bulk128.example`	trust-warm	0/6	6/6
clean	1232	`bulk192.example`	cold	0/6	3/6
clean	1232	`bulk192.example`	trust-warm	0/6	6/6
harsh	4096	`bulk128.example`	cold	0/6	0/6
harsh	4096	`bulk128.example`	trust-warm	0/6	5/6
harsh	4096	`bulk192.example`	cold	0/6	4/6
harsh	4096	`bulk192.example`	trust-warm	0/6	5/6

The EDNS-4096 authoritative window is real within the tested Falcon-512 region, and its boundary is signature-family sensitive.

VI. DISCUSSION

The experiments characterize a transport frontier governed by five variables in the evaluated artifacts: EDNS size, effective response size, network profile, cache phase, and signature family. Separating those variables gives a clearer reading of the comparison.

SL-DNSSEC and TurboDNS induce different transport regimes. SL-DNSSEC consistently exposes a more UDP-oriented first resolver-visible leg, reflecting a structural property of the design. In the same environment, TurboDNS obtains an EDNS-4096 authoritative DNS-layer completion window under Falcon-512. That window becomes most relevant in the large-answer rows, where the evaluated SL-DNSSEC artifact and configuration fail to sustain validating completion.

Local authoritative completion and end-to-end validating completion measure different parts of the system. The packet captures show the split directly. A local authoritative DNS-layer UDP-complete region for TurboDNS can coexist with more TCP frames, more retransmissions, or a heavier first visible leg in the validating path. The three measurement

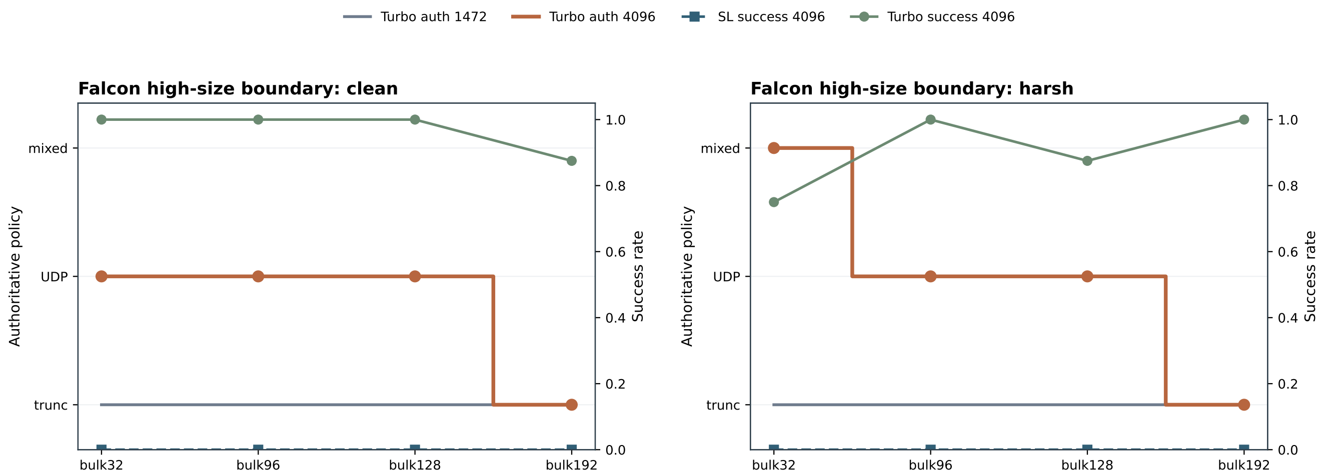

Fig. 5. Falcon-512 high-size boundary. Under EDNS 4096, TurboDNS remains DNS-layer authoritative UDP-complete through `bulk128.example` and loses that window at `bulk192.example`; EDNS 1472 does not buy the same transition. DNS-layer completion does not imply fragmentation-free delivery.

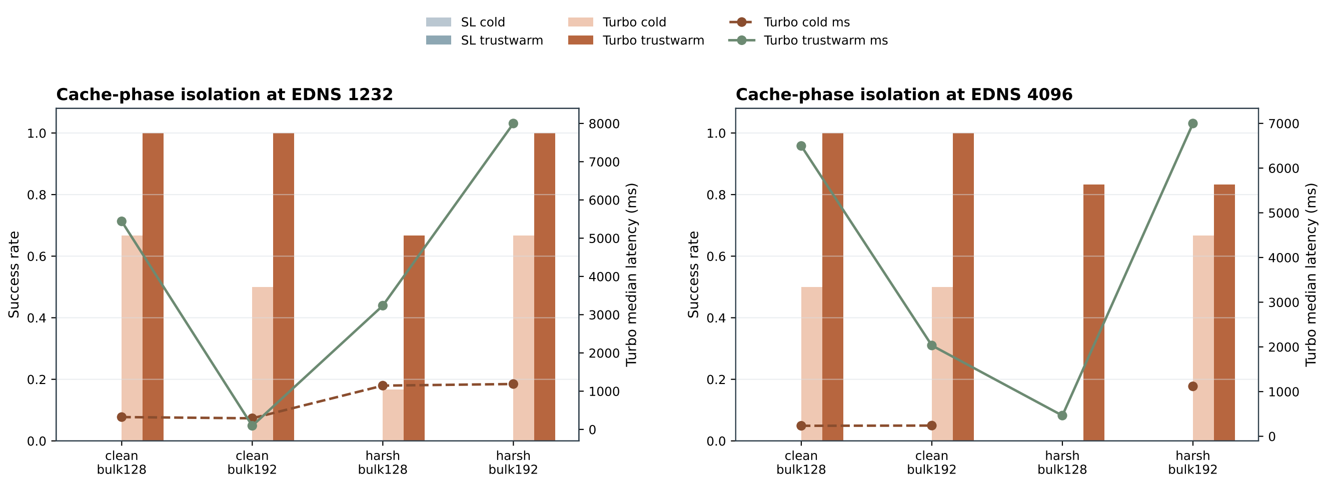

Fig. 6. Isolated cold versus trust-warm behavior near the large-answer boundary. Trust-material warmup materially improves TurboDNS robustness in several rows, while SL-DNSSEC remains without sustained validating completion in the same regime. Medians are conditional on successful validating completion.

planes therefore expose distinct costs: authoritative response completion, resolver-visible behavior, and full validating completion. A benchmark limited to the authoritative server would miss a substantial part of the operational cost.

EDNS 1472 occupies a weak middle ground in this setting. Across the core rows and the size-boundary rows, 1472 behaves closer to 1232 than to 4096 on the authoritative path. It fails to produce the Falcon-512 local window and often inherits the fragility of the smaller regime without reaching the larger-regime transition.

For large signed `A` answers, TurboDNS turns a costly cold first access into a more reliable trust-warm access. The effect is strongest in robustness, while latency varies by row: some warm rows become much faster, and others remain expensive even as success improves. SL-DNSSEC remains without sustained validating completion in the same regime. The path to trust material is therefore part of the deployment story, not background plumbing.

Under Falcon-512, the data support a useful EDNS-4096 authoritative DNS-layer UDP-complete region for TurboDNS. Under Dilithium2, the tested range is truncated from the first point. The frontier is therefore partly algorithmic: the fallback design interacts with the signature family carried by the signed path.

For deployment, a practitioner evaluating the designs should consider the EDNS values available in the target environment, the size of the DNSKEY and signed-answer paths under the intended signature family, the observation plane that carries the main operational constraint, and the expected frequency of cold validating paths relative to trust-warmed ones. These factors are more informative than a generic latency ranking, since the same mechanism can appear favorable in one plane and costly in another.

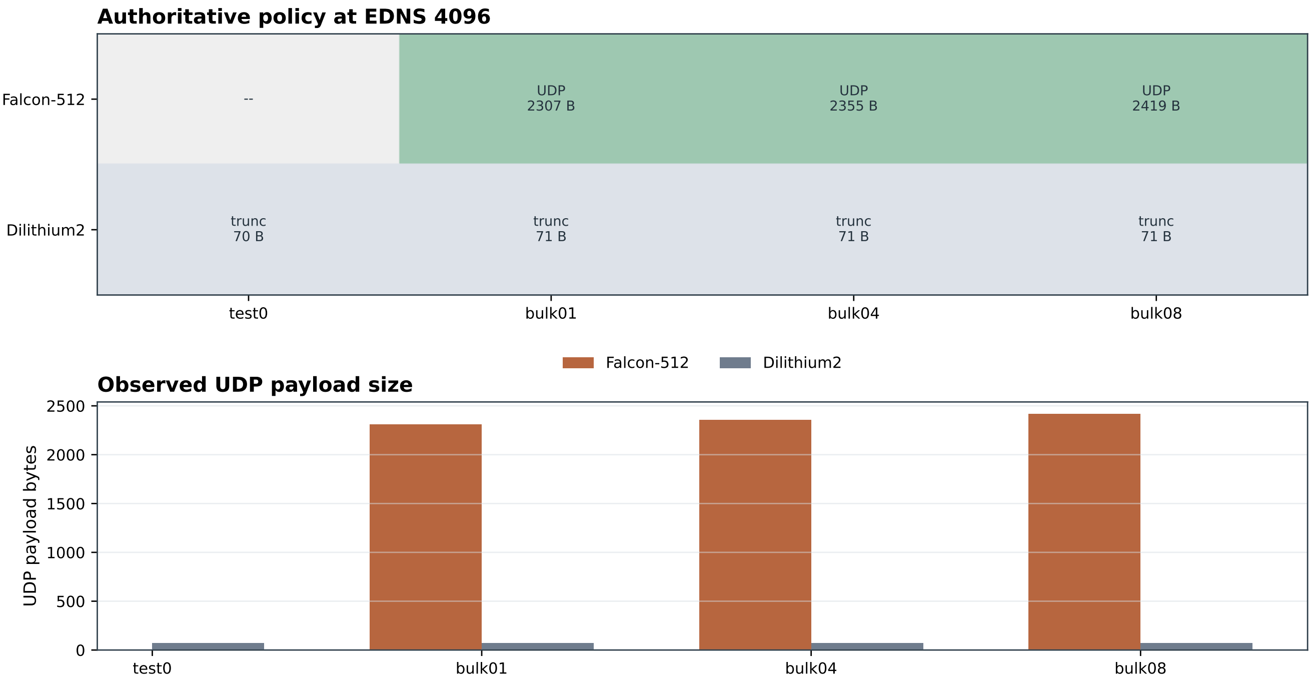

Fig. 7. Signature-family sensitivity under EDNS 4096. Falcon-512 preserves a non-empty Turbo DNS-layer authoritative UDP-complete window, whereas Dilithium2 is truncated from the first tested point. The Dilithium2 UDP payloads shown here are TC-only replies.

A. *Threats to validity*

The scope of the evaluation is a controlled local harness with two public artifacts. The `bulkNN` ladder is a controlled payload ladder, with no deployment-frequency model for common RRsets. The main `udp_complete` labels describe DNS-layer behavior; EDNS 4096 can still produce replies beyond the path MTU. Focused campaigns such as cache isolation and signature sensitivity use 4–8 repetitions per cell and support micro-study interpretations, with lower confidence for global rate estimates. The large-answer SL-DNSSEC failures reported here are artifact-level operational outcomes for the evaluated implementation and configuration.

VII. Conclusion

We evaluated TurboDNS and SL-DNSSEC by running their public artifacts in a common four-node validating DNSSEC environment and measuring authoritative completion, first resolver-visible behavior, and full validating completion with repeated probes and packet captures. The resulting measurements expose the transport regimes induced by each design under shared network profiles, EDNS budgets, response-size ladders, cache phases, and signature families.

In the core Falcon-512 experiments, EDNS 4096 moves TurboDNS authoritative DNSKEY delivery into DNS-layer UDP completion, while SL-DNSSEC remains truncated on that path. The same split appears with inflated signed `A` responses: under Falcon-512, TurboDNS keeps the EDNS-4096 authoritative window through `bulk128.example` and loses it at `bulk192.example`. Resolver-side captures show the remaining end-to-end validation cost in that region, including TCP frames and retransmissions. Trust-material warmup improves TurboDNS robustness near the large-answer boundary, while the evaluated SL-DNSSEC artifact remains without sustained validating completion in those rows. The signature-family micro-study bounds the result further: the Falcon-512 window exists in a non-empty tested region, and the tested Dilithium2 range is truncated from the first point.

For post-quantum DNSSEC mechanisms, deployment should be evaluated through the transport regime induced across the validating path, conditioned on EDNS size, effective response size, network profile, cache state, and signature family. The measured frontier exposes that regime and provides a comparative basis for deciding where each design offers a practical advantage.

VIII. Ethics Considerations

The study evaluates public research artifacts in an isolated local environment using synthetic DNS zones and controlled network conditions. It does not involve human subjects, collection of user data, or interaction with third-party infrastructure. No new vulnerabilities were discovered in production systems. The experiments were designed to compare transport behavior in a contained testbed and therefore did not raise additional disclosure obligations.

References

[1] J. L. S. Damas, M. Graff, and P. A. Vixie, “Extension mechanisms for DNS (EDNS(0)),” RFC 6891, Apr. 2013. [Online]. Available: https://www.rfc-editor.org/info/rfc6891

[2] R. Bonica, F. Baker, G. Huston, B. Hinden, O. Trøan, and F. Gont, "IP fragmentation considered fragile," RFC 8900, Sep. 2020. [Online]. Available: https://www.rfc-editor.org/info/rfc8900
[3] G. C. M. Moura, M. Müller, M. Davids, M. Wullink, and C. Hesselman, "Fragmentation, truncation, and timeouts: Are large DNS messages falling to bits?" in *Passive and Active Measurement*, ser. Lecture Notes in Computer Science, O. Hohlfeld, A. Lutu, and D. Levin, Eds., vol. 12671. Cham: Springer, 2021, pp. 460–477.
[4] M. Müller, J. de Jong, M. van Heesch, B. Overeinder, and R. van Rijswijk-Deij, "Retrofitting post-quantum cryptography in internet protocols: a case study of DNSSEC," *ACM SIGCOMM Computer Communication Review*, vol. 50, no. 4, pp. 49–57, Oct. 2020.
[5] A. S. Rawat and M. P. Jhanwar, "Post-quantum DNSSEC with faster TCP fallbacks," in *Progress in Cryptology – INDOCRYPT 2024*, ser. Lecture Notes in Computer Science, S. Mukhopadhyay and P. Stănică, Eds., vol. 15496. Cham: Springer, 2025, pp. 212–236.
[6] ——, "Quantum-safe signatureless DNSSEC," in *Proceedings of the 20th ACM Asia Conference on Computer and Communications Security*, ser. ASIA CCS '25. New York, NY, USA: Association for Computing Machinery, 2025, pp. 267–282.
[7] ——, "On the security of SL-DNSSEC," Cryptology ePrint Archive, Paper 2025/1770, 2025. [Online]. Available: https://eprint.iacr.org/2025/1770
[8] R. Arends, R. Austein, M. Larson, D. Massey, and S. Rose, "DNS security introduction and requirements," RFC 4033, Mar. 2005. [Online]. Available: https://www.rfc-editor.org/info/rfc4033
[9] ——, "Resource records for the DNS security extensions," RFC 4034, Mar. 2005. [Online]. Available: https://www.rfc-editor.org/info/rfc4034
[10] ——, "Protocol modifications for the DNS security extensions," RFC 4035, Mar. 2005. [Online]. Available: https://www.rfc-editor.org/info/rfc4035
[11] G. van den Broek, R. M. van Rijswijk-Deij, A. Sperotto, and A. Pras, "DNSSEC meets real world: dealing with unreachability caused by fragmentation," *IEEE Communications Magazine*, vol. 52, no. 4, pp. 154–160, Apr. 2014.
[12] J. Goertzen and D. Stebila, "Post-quantum signatures in DNSSEC via request-based fragmentation," in *Post-Quantum Cryptography*, ser. Lecture Notes in Computer Science, T. Johansson and D. Smith-Tone, Eds., vol. 14154. Cham: Springer, 2023, pp. 535–564.
[13] A. S. Rawat and M. P. Jhanwar, "Post-quantum DNSSEC over UDP via QNAME-based fragmentation," in *Security, Privacy, and Applied Cryptography Engineering*, ser. Lecture Notes in Computer Science, F. Regazzoni, B. Mazumdar, and S. Parameswaran, Eds., vol. 14412. Cham: Springer, 2024, pp. 66–85.

Appendix A
Additional Experimental Details

A. Stress and sensitivity branches

The appendix reports two auxiliary experiment branches.

The first branch is a 512-byte EDNS stress campaign covering both systems under a constrained payload setting.

The second branch is an MTU sensitivity study over 1500 and 1280 bytes, covering robustness, retransmission burden, and tail behavior for the evaluated artifacts.

B. Artifact contents

The accompanying artifact contains the public prototype baselines, the build and execution scripts used to place them in a common environment, the campaign runners, and the packet captures.

TABLE VII
512-byte EDNS stress summary.

Net.	EDNS	SL succ.	Turbo succ.	SL auth. DNSKEY	Turbo auth. DNSKEY	Med. A SL/T
clean	512	0.875	1.000	trunc.	trunc.	1023.5/3137.5